

* Rank 0 tensor.

$$[T]$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} T_x & 0 & T_z \end{bmatrix}$$

$$T_x = \mathbb{T}_x^{(1)}$$

$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & 0 & T_{xz} \\ 0 & T_{yy} & 0 \\ T_{xz} & 0 & T_{zz} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$

$$T_{xz} = \mathbb{T}_{xz}^{(1)}$$

$$T_{yy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$

$$T_{zz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & T_{xz} \\ 0 & 0 & 0 \\ -T_{xz} & 0 & 0 \end{bmatrix}$$

$$T_{xz} = -\mathbb{M}_y^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxxx} & 0 & T_{xxzz} \\ T_{yyyx} & 0 & T_{yyyz} \\ T_{zzzx} & 0 & T_{zzzz} \\ 0 & T_{yyyz} & 0 \\ T_{xxzz} & 0 & T_{zzzx} \\ 0 & T_{yyyx} & 0 \end{bmatrix}$$

$$T_{xxx} = 3\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)}$$

$$T_{xxz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\beta(1)}$$

$$T_{yyx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\beta(1)}$$

$$T_{yyz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{zzx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\beta(1)}$$

$$T_{zzz} = 3\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ T_{yyx} & 0 & T_{yyz} \\ T_{zzx} & 0 & 0 \\ 0 & -\frac{T_{yyz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & -\frac{T_{zzx}}{2} \\ 0 & -\frac{T_{yyx}}{2} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxz} &= 2\mathbb{M}_{xy}^{(1)} + 2\mathbb{T}_z^{(2)} \\
T_{yyx} &= 2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)} \\
T_{yyz} &= -2\mathbb{M}_{xy}^{(1)} + 2\mathbb{T}_z^{(2)} \\
T_{zzx} &= -2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)} \\
\frac{T_{yyz}}{2} &= -\mathbb{M}_{xy}^{(1)} + \mathbb{T}_z^{(2)} \\
\frac{T_{xxz}}{2} &= \mathbb{M}_{xy}^{(1)} + \mathbb{T}_z^{(2)} \\
\frac{T_{zzx}}{2} &= -\mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)} \\
\frac{T_{yyx}}{2} &= \mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}
\end{aligned}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & T_{yzy} & 0 \\ T_{zxx} & 0 & T_{zzx} \\ 0 & T_{xyy} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{yzy} &= \mathbb{M}_{xy}^{(2)} + \mathbb{T}_z^{(3)} \\
T_{zxx} &= \mathbb{M}_{xy}^{(2)} - \mathbb{T}_z^{(3)} \\
T_{zzx} &= \mathbb{M}_{yz}^{(2)} + \mathbb{T}_x^{(3)} \\
T_{xyy} &= \mathbb{M}_{yz}^{(2)} - \mathbb{T}_x^{(3)}
\end{aligned}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & T_{xxzx} & 0 \\ T_{xxyy} & T_{yyyy} & T_{yyzz} & 0 & T_{yyzx} & 0 \\ T_{xxzz} & T_{yyzz} & T_{zzzz} & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & T_{yzxy} \\ T_{xxzx} & T_{yyzx} & T_{zzzx} & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & T_{yzxy} & 0 & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{xxxx} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)} \\
T_{xxyy} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} \\
T_{xxzz} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)} \\
T_{xxzx} &= -\mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_y^{\beta(1)} \\
T_{yyyy} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)} \\
T_{yyzz} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)} \\
T_{yyzx} &= \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_y^{\beta(1)} \\
T_{zzzz} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)} \\
T_{zzzx} &= \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_y^{\beta(1)} \\
T_{yzyz} &= \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)} \\
T_{yzxy} &= \mathbb{T}_{xz}^{(2)} + 2\mathbb{T}_y^{\beta(1)} \\
T_{zzxx} &= \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)} \\
T_{xyxy} &= \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)}
\end{aligned}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & T_{xxyy} & T_{xxzz} & 0 & T_{xxzx} & 0 \\ -T_{xxyy} & 0 & T_{yyzz} & 0 & T_{yyzx} & 0 \\ -T_{xxzz} & -T_{yyzz} & 0 & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxy} \\ -T_{xxzx} & -T_{yyzx} & -T_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xyyy} &= 4\mathbb{M}_{xyz}^{(1)} - 2\mathbb{T}_v^{(3)} \\
T_{xxzz} &= -4\mathbb{M}_{xyz}^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(3)} \\
T_{xxxz} &= -2\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(3)} \\
T_{yyzz} &= 4\mathbb{M}_{xyz}^{(1)} + 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(3)} \\
T_{yyzx} &= -4\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(3)} \\
T_{zzzx} &= 2\mathbb{M}_y^{(1)} - 2\mathbb{M}_y^{\alpha(1)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(3)} \\
T_{yzxy} &= \mathbb{M}_y^{(1)} + 4\mathbb{M}_y^{\alpha(1)}
\end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & 0 & T_{yzxy} \\ 0 & T_{zzzx} & 0 \\ T_{yzxy} & 0 & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{yzyz} &= \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(4)} \\
T_{yzxy} &= 2\mathbb{T}_{xz}^{(4)} \\
T_{zzzx} &= \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 2\mathbb{T}_v^{(4)} \\
T_{xyxy} &= \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}
\end{aligned}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & T_{yzxy} \\ 0 & 0 & 0 \\ -T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$T_{yzxy} = \mathbb{M}_y^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxzx} & 0 \\ 0 & T_{yyzx} & 0 \\ 0 & T_{zzzx} & 0 \\ T_{yzyz} & 0 & T_{yzxy} \\ 0 & T_{zzzx} & 0 \\ T_{xyyz} & 0 & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{xxzx} &= \mathbb{M}_y^{(3)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{xz}^{(5)} \\
T_{yyzx} &= \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{(4)} + 2\mathbb{M}_y^{\alpha(2)} \\
T_{zzzx} &= \mathbb{M}_y^{(3)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} - 2\mathbb{T}_{xz}^{(5)} \\
T_{yzyz} &= \mathbb{M}_{xyz}^{(2)} - 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(5)} \\
T_{yzxy} &= \mathbb{M}_y^{(4)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(5)} \\
T_{zzzx} &= \mathbb{M}_{xyz}^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(5)} \\
T_{xyyz} &= \mathbb{M}_y^{(4)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{xz}^{(5)} \\
T_{xyxy} &= \mathbb{M}_{xyz}^{(2)} + 2\mathbb{T}_v^{(5)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & T_{yzxy} \\ T_{zzxx} & T_{zxyy} & T_{zzzz} & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & T_{xyyz} & 0 & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{yzyz} &= \mathbb{M}_{xyz}^{(3)} - 3\mathbb{T}_u^{(6)} - \mathbb{T}_v^{(6)} \\
T_{yzxy} &= \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} - \mathbb{T}_{xz}^{(6)} \\
T_{zxxx} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(6)} \\
T_{zxyy} &= \mathbb{M}_y^{(5)} + 2\mathbb{M}_y^{(6)} + 2\mathbb{M}_y^{\alpha(3)} \\
T_{zxxx} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(6)} \\
T_{zxzx} &= \mathbb{M}_{xyz}^{(3)} + 3\mathbb{T}_u^{(6)} - \mathbb{T}_v^{(6)} \\
T_{xyyz} &= \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} + \mathbb{T}_{xz}^{(6)} \\
T_{xyxy} &= \mathbb{M}_{xyz}^{(3)} + 2\mathbb{T}_v^{(6)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & T_{xxzx} & 0 & 0 & T_{xxxx} & 0 \\
T_{yyxx} & T_{yyyy} & T_{yyzz} & 0 & T_{yyzx} & 0 & 0 & T_{yyxz} & 0 \\
T_{zzxx} & T_{zzyy} & T_{zzzz} & 0 & T_{zzzx} & 0 & 0 & T_{zzxz} & 0 \\
0 & 0 & 0 & T_{yzyz} & 0 & T_{yzxy} & T_{yzzy} & 0 & T_{yzyx} \\
T_{zxxx} & T_{zxyy} & T_{zxzz} & 0 & T_{zxzx} & 0 & 0 & T_{zxzx} & 0 \\
0 & 0 & 0 & T_{xyyz} & 0 & T_{xyxy} & T_{xyzy} & 0 & T_{xyyx}
\end{bmatrix}$$

$$\begin{aligned}
T_{xxxx} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)} \\
T_{xxyy} &= 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 2\mathbb{T}_v^{(3)} \\
T_{xxzz} &= -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} - \mathbb{T}_v^{(3)} \\
T_{xxxz} &= -2\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)} \\
T_{xxxz} &= -2\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)} \\
T_{yyxx} &= -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} + 2\mathbb{T}_v^{(3)} \\
T_{yyyy} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)} \\
T_{yyzz} &= 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)} \\
T_{yyzx} &= \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{(4)} + 2\mathbb{M}_y^{\alpha(2)} - 4\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_y^{\beta(1)} \\
T_{yyxz} &= -\mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{(4)} - 2\mathbb{M}_y^{\alpha(2)} - 4\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_y^{\beta(1)} \\
T_{zzxx} &= 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)} \\
T_{zzyy} &= -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} - \mathbb{T}_v^{(3)} \\
T_{zzzz} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)} \\
T_{zzzx} &= 2\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)} \\
T_{zzzx} &= 2\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)} \\
T_{yzyz} &= \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} - \mathbb{T}_v^{(5)} \\
T_{yzxy} &= \mathbb{M}_y^{(1)} + \mathbb{M}_y^{(4)} + 4\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)} \\
T_{yzzy} &= -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)} \\
T_{yzyx} &= \mathbb{M}_y^{(1)} - \mathbb{M}_y^{(4)} + 4\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)} \\
T_{zxxx} &= 2\mathbb{M}_y^{(1)} - 2\mathbb{M}_y^{\alpha(1)} - 2\mathbb{M}_y^{\beta(1)} - \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)} - \mathbb{T}_y^{\beta(1)} \\
T_{zxyy} &= 4\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(1)} - \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_y^{\beta(1)} \\
T_{zxzx} &= -2\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)} - 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)} - \mathbb{T}_y^{\beta(1)} \\
T_{zxzx} &= \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} - \mathbb{T}_v^{(5)} \\
T_{zxzx} &= -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)} \\
T_{xyyz} &= -\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(4)} - 4\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)} \\
T_{xyxy} &= \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(5)} \\
T_{xyzy} &= -\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(4)} - 4\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)} \\
T_{xyyx} &= -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(5)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & T_{zyyz} & 0 & T_{yzxy} & T_{yzzy} & 0 & T_{zyyx} \\ T_{zxxx} & T_{zxyy} & T_{zxxz} & 0 & T_{zxzx} & 0 & 0 & T_{zxxz} & 0 \\ 0 & 0 & 0 & T_{xyyz} & 0 & T_{xyxy} & T_{xyzy} & 0 & T_{xyyx} \end{bmatrix}$$

$$\begin{aligned} T_{zyyz} &= \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} + 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\ T_{yzxy} &= \mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(4)} - \mathbb{T}_{xz}^{(6)} \\ T_{yzzy} &= \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} - 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\ T_{zyyx} &= -\mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(4)} - \mathbb{T}_{xz}^{(6)} \\ T_{zxxx} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(6)} \\ T_{zxyy} &= \mathbb{M}_y^{(5)} + 2\mathbb{M}_y^{(6)} + 2\mathbb{M}_y^{\alpha(3)} \\ T_{zxzx} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(6)} \\ T_{zxzx} &= \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} + 3\mathbb{T}_u^{(6)} - 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\ T_{zxzx} &= \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} + 3\mathbb{T}_u^{(6)} + 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\ T_{xyyz} &= -\mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(4)} + \mathbb{T}_{xz}^{(6)} \\ T_{xyxy} &= \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(6)} \\ T_{xyzy} &= \mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(4)} + \mathbb{T}_{xz}^{(6)} \\ T_{xyyx} &= \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} - 4\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(6)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & 0 & T_{xxxx} \\ 0 & T_{yyyy} & 0 \\ T_{zzzz} & 0 & T_{zzzz} \\ T_{yyzx} & 0 & T_{yyzz} \\ T_{zzxx} & 0 & T_{zzxz} \\ 0 & T_{xxyy} & 0 \\ 0 & T_{yzzy} & 0 \\ T_{zzxx} & 0 & T_{zzxz} \\ T_{xxyy} + T_{yyzz} - T_{yzzy} - T_{zxzx} + T_{zzxx} & 0 & T_{xyyz} \\ 0 & T_{xyyz} + \frac{T_{zzxx}}{2} - \frac{T_{zzxz}}{2} & 0 \end{bmatrix}$$

$$\begin{aligned} T_{xxxx} &= 3\mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)} \\ T_{xxxx} &= -\mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)} \\ T_{yyyy} &= 3\mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)} \\ T_{zzzz} &= \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)} \\ T_{zzzz} &= 3\mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)} \\ T_{yyzx} &= \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_y^{\beta(1)} \\ T_{yyzz} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)} \\ T_{zzxx} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)} \\ T_{zzxz} &= \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)} \\ T_{xxyy} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 2\mathbb{T}_v^{(3)} \\ T_{yzzy} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)} \\ T_{zxxx} &= -\mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)} - \mathbb{T}_y^{\beta(1)} \\ T_{zxzx} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)} \\ T_{xyyz} &= \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)} \end{aligned}$$